Dark Horse Prototype 1: Check-in Butler Robot Drone

Aryan Sood, Himansu Pathak, Junyu Zhou
Ilham, Jolie, Gavin Tey, Lim Soong En

Stanford University

Critical Experience Prototype

We wanted to tackle the check-in experience from this **completely** different angle

This meant designing various parts for an end-to-end check-in flow

What is Luxury?

1. What is most socially desirable but not accessible to all

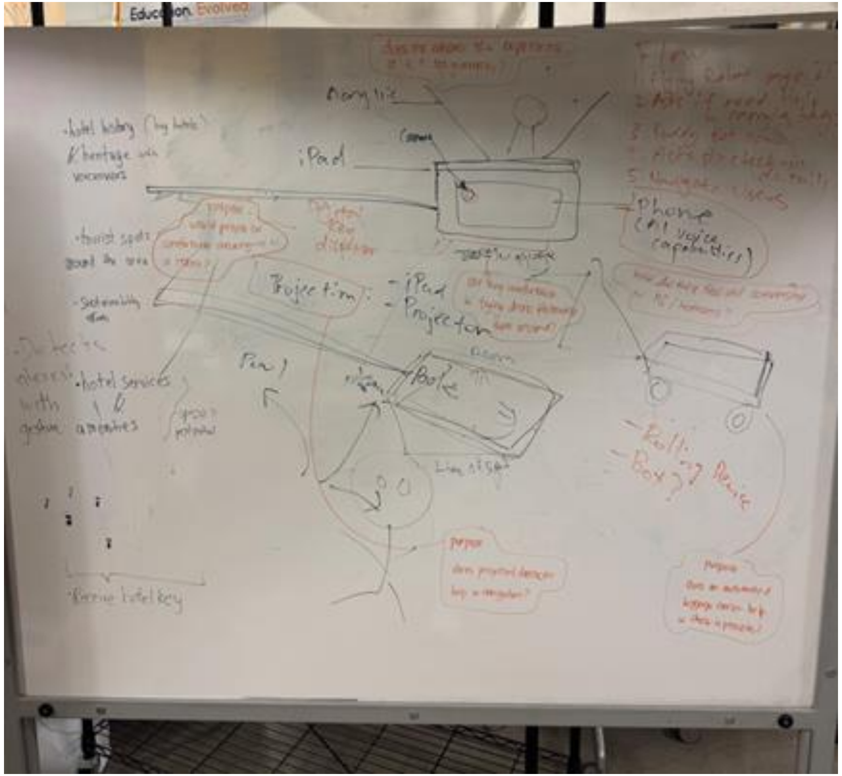
2. Luxury is needlessly expensive

3. Outstanding products with emotional qualities

The Core Idea (Recap)

The Experience:

1. Guest arrives; a small, friendly drone (The "AeroHost") flies and descends to greet them.
2. Facial recognition confirms identity instantly.
3. The robot says, "Welcome, [Name], follow me.", and asks if the guest needs help with their luggage
    i. If so, it calls a buddy ground robot that is able to roll the luggage around
4. It physically leads them through the lobby, to the elevator, and to their room.
5. The robot is able to answer any questions.



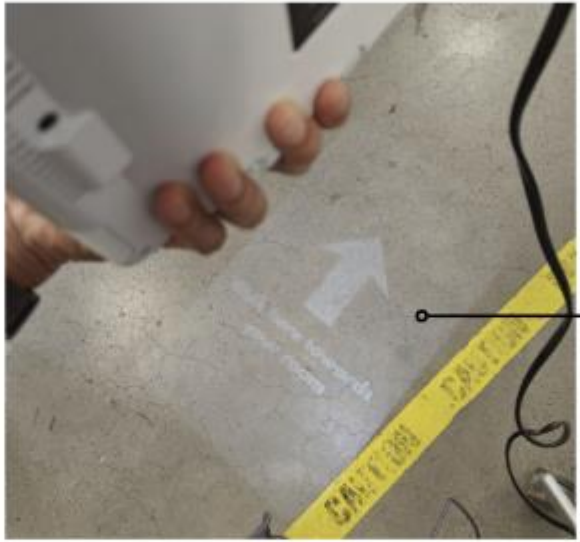
Our Prototyping Plan (v1)

1. The Drone Body
   a. AI conversational capabilities (Custom GPT on phone)
   b. Holographic projection (iPad)
   c. Pole attachment for flying simulation
2. The Buddy Luggage Bot
   a. A rolling device to be pulled manually
3. Projection
   a. A small projector adjusted manually to display navigation & check-in code

hologram

AI conversational assistant

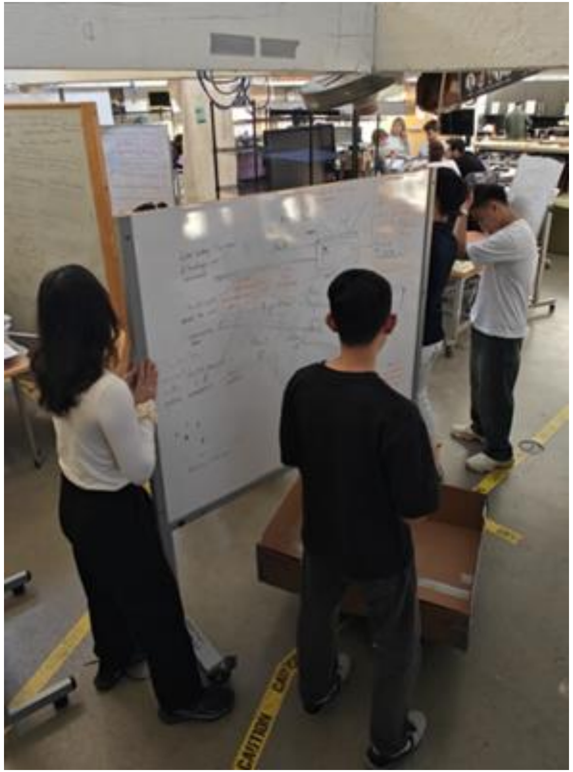
Luggage Bot

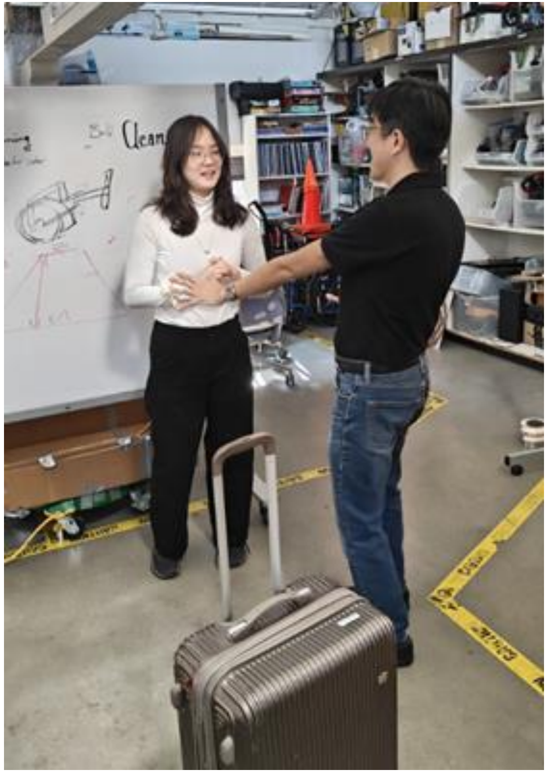


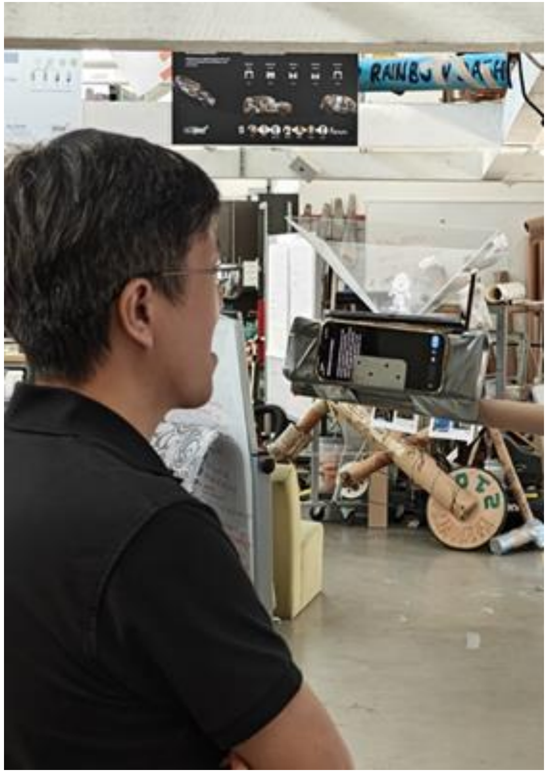
Navigation

Check-In QR

Our Wizard-Of-Oz Experience

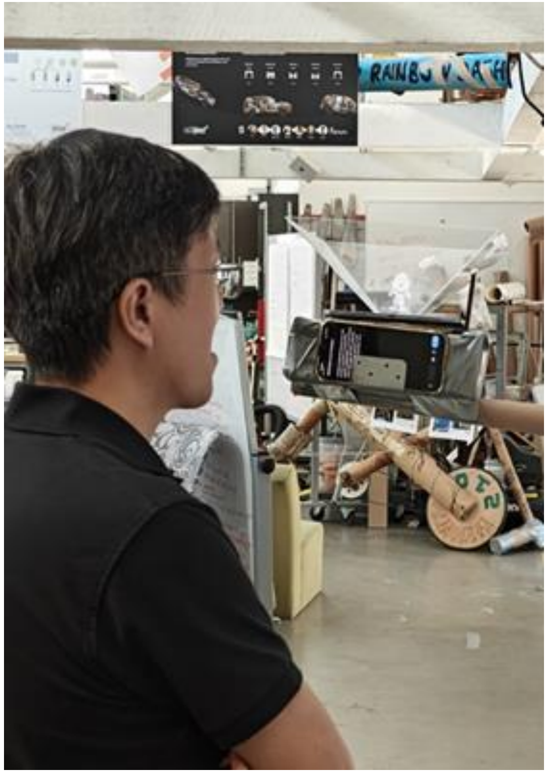






Our Wizard-Of-Oz Experience

When conversing with the bot, the bot is able to greet guests and coordinate luggage assistance by calling the assistant luggage bot



Our Wizard-Of-Oz Experience

It will then call the luggage bot and guests would need to place the luggage on the trolley

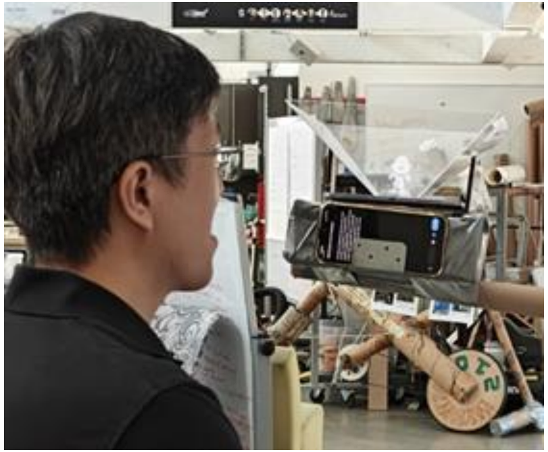
Our Wizard-Of-Oz Experience

The bot then guides guests through a full self-service check-in process, this includes

(i) scanning the passport

(ii) paying deposit

(iii) issue mobile keys through QR



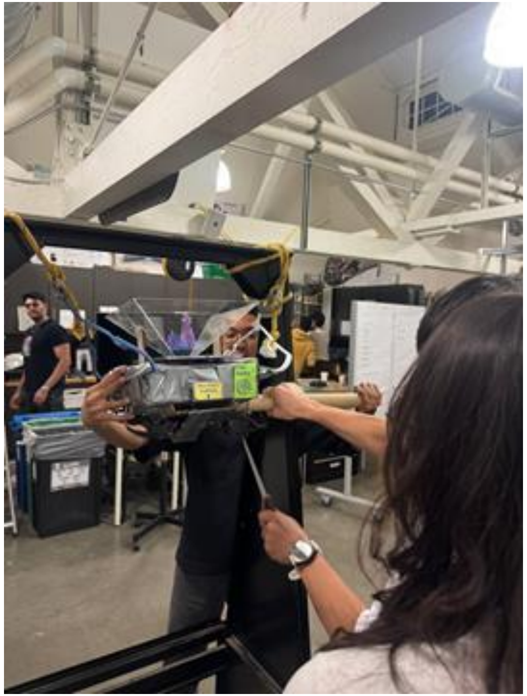
Our Wizard-Of-Oz Experience

The bot then guides guests to the room while conversing naturally with the guests

Key points that we were looking out for

- Does the hologram enhance the experience / personalisation, or is it just a gimmick?
- Are users comfortable walking with a drone and luggage bot?
- Are guests able to naturally understand how to speak to and interact with a flying robot without a screen-based interface?
- Do guests feel confident that their belongings are secure when luggage is handed off to the simulated “Buddy Bot” without direct human supervision?
- Is the “Digital Key Dispenser” interaction intuitive and seamless for guests?
- Does the reduction of visible human staff increase feelings of delight and premium service, or does it create discomfort and a sense of being unattended?

Test Feedback and Iterative Enhancements

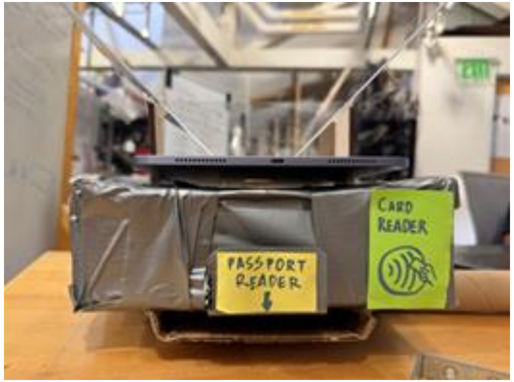


Feedback and Iterative Enhancements: Drone

01

"Unsure of where to scan passport/card"
"The drone could look more like a drone"

Initially, the AI only instructed the guest to scan passport/card, but there was no physical display or clues on how to do so



Feedback and Iterative Enhancements: Drone

02

“It doesn’t really feel like I’m interacting with a **hovering** drone”

Feedback and Iterative Enhancements: Robot Interactions

03

"The distance of the drone from me is good, but it feels like the luggage bot was too far behind at times"

From a UX standpoint, we underestimated the importance of the distance of bots from the guest

Feedback and Iterative Enhancements: Conversational AI

04

"The drone AI stopped talking to me once the check-in was complete, but perhaps it could make small talk"

We enhanced the human touch by making the AI act like a personal butler EVEN after the check-in is complete

Feedback and Iterative Enhancements: Luggage Bot

05

"Having the guest carry their luggage onto the luggage bot does not seem luxurious. It is quite a lot of effort."

"Is the robot going to help me or am I supposed to?"

Demo

Future Enhancements: Luggage Bot

Idea for potential implementation of the luggage bot

Future Enhancements: Luggage Bot

"Having the guest carry their luggage onto the luggage bot does not seem luxurious. It is quite a lot of effort."

"Is the robot going to help me or am I supposed to?"

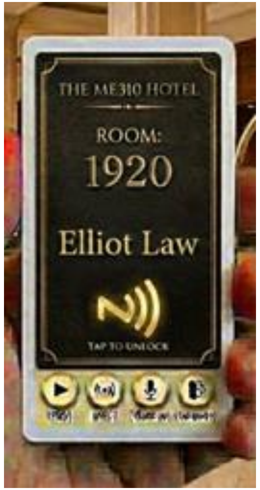
Future Enhancements:

Idea for potential implementation of other form factor of the AI assistant

Future Enhancements:

“Does it need to float? Not necessarily.”

“If we can integrate the character doing various actions etc, and the voice, it’ll be really good. I like to see a very happy snoopy jump around.”



Results (5-point scale)

Luxury	3.6
Security and Trust	4
Social Comfort	4.4
Cognitive Clarity	4.2

Key learning points

- No matter how “cool” the tech is, we have to be mindful to make the experience as seamless and low-effort as possible

- We cannot rely solely on screens, voice or holograms to guide physical actions. High-tech interactions need low-tech “anchors” (words, instructions, displays)

- A check-in solution cannot just “end” when the key is issued - it must bridge the gap to the next step (finding elevator, entering room, or even throughout the stay)

What we learnt about luxury

1. Luxury is not the tech; it's the feeling of being personally cared for
   - Across everyone's comments, luxury was *not* automatically associated with robots, drones, holograms, or AI
1. Cognitive ease is a core component of luxury
   - Confusion destroys luxury
1. Convenience as a core signal of luxury
   - From the feedback, luxury consistently aligned with how little the guest had to decide, or do